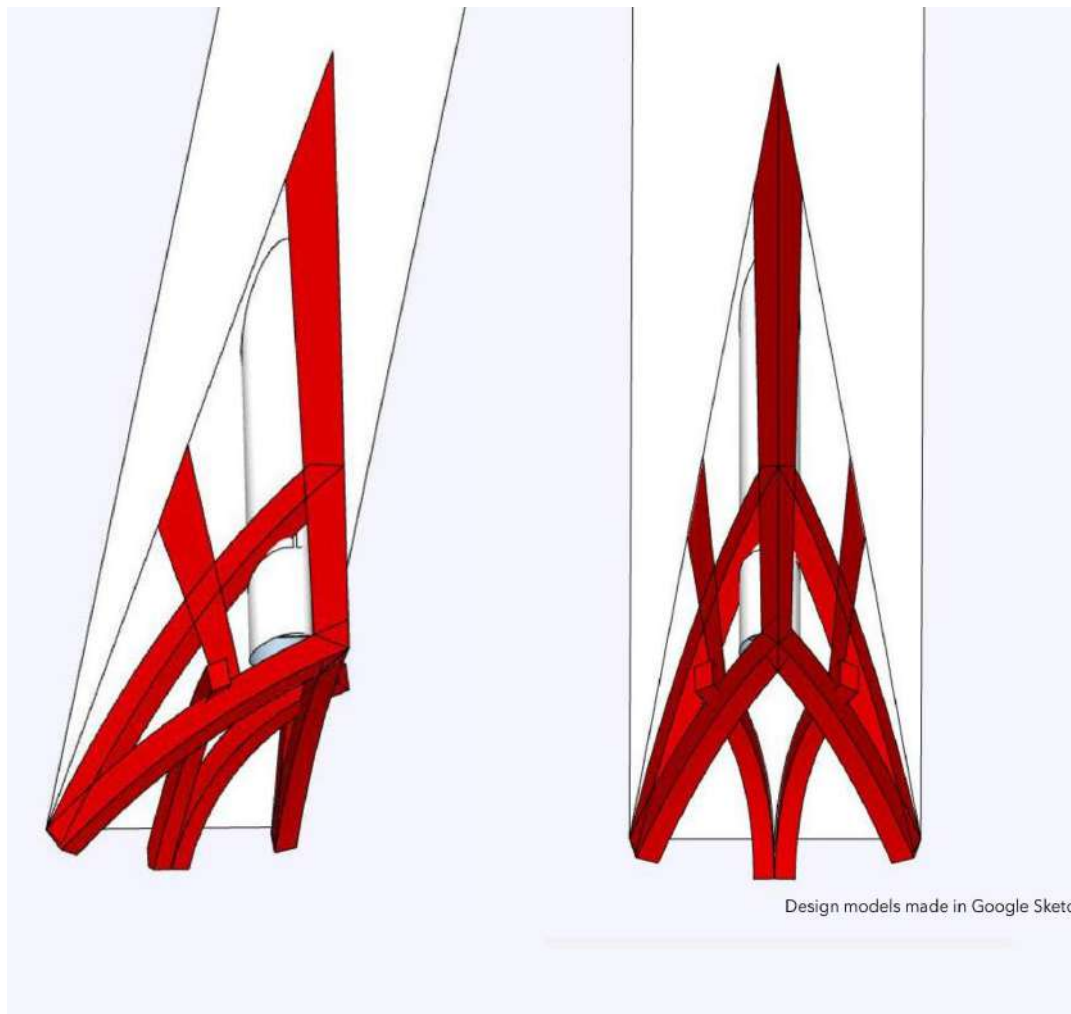




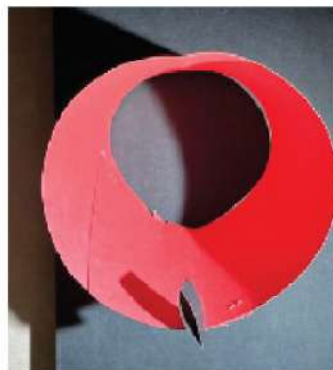
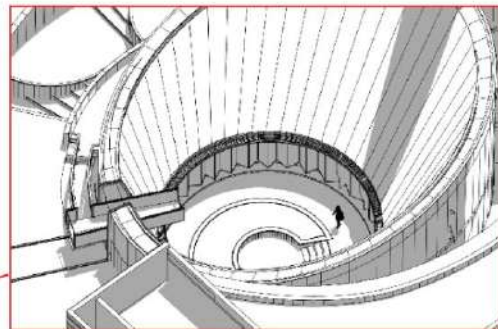
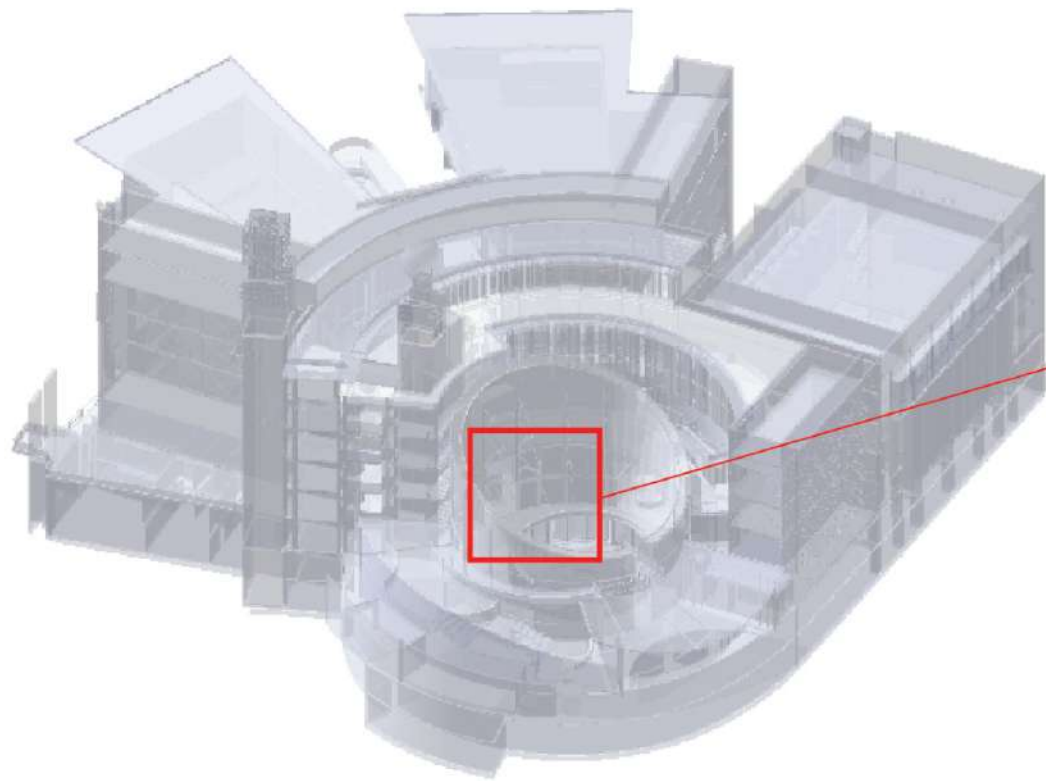
Custom Solar Light Pipe for the Ark Museum, Vadodara

Context

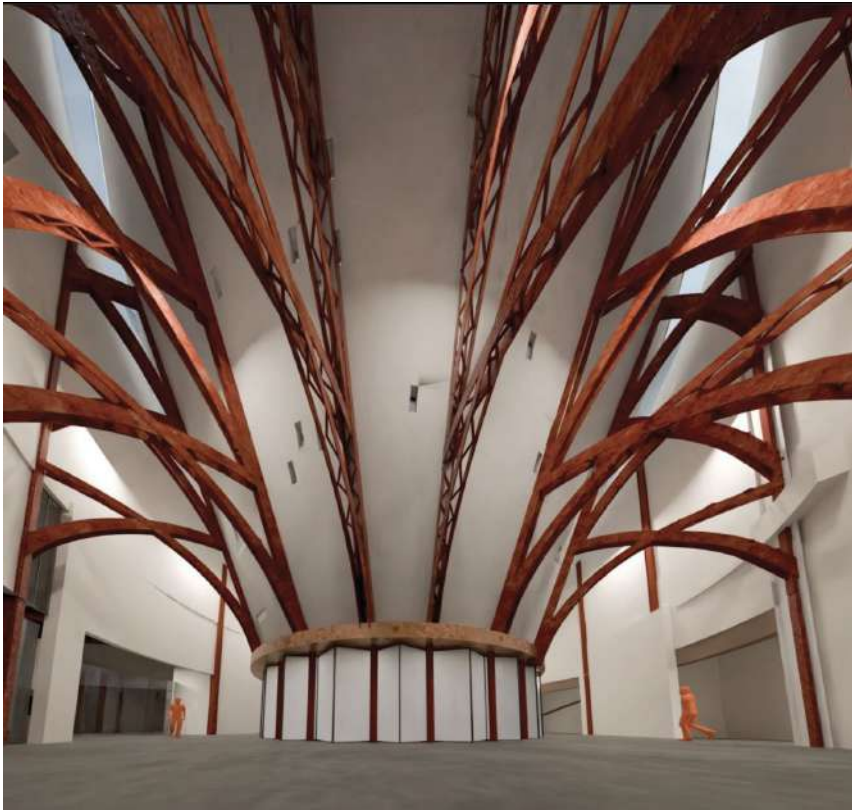
A lighting solution expanding on the concept of a solar pipe as a bespoke fixture design for the Ark Foundation Museum, Vadodara

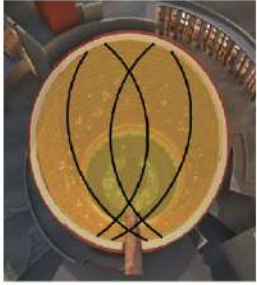


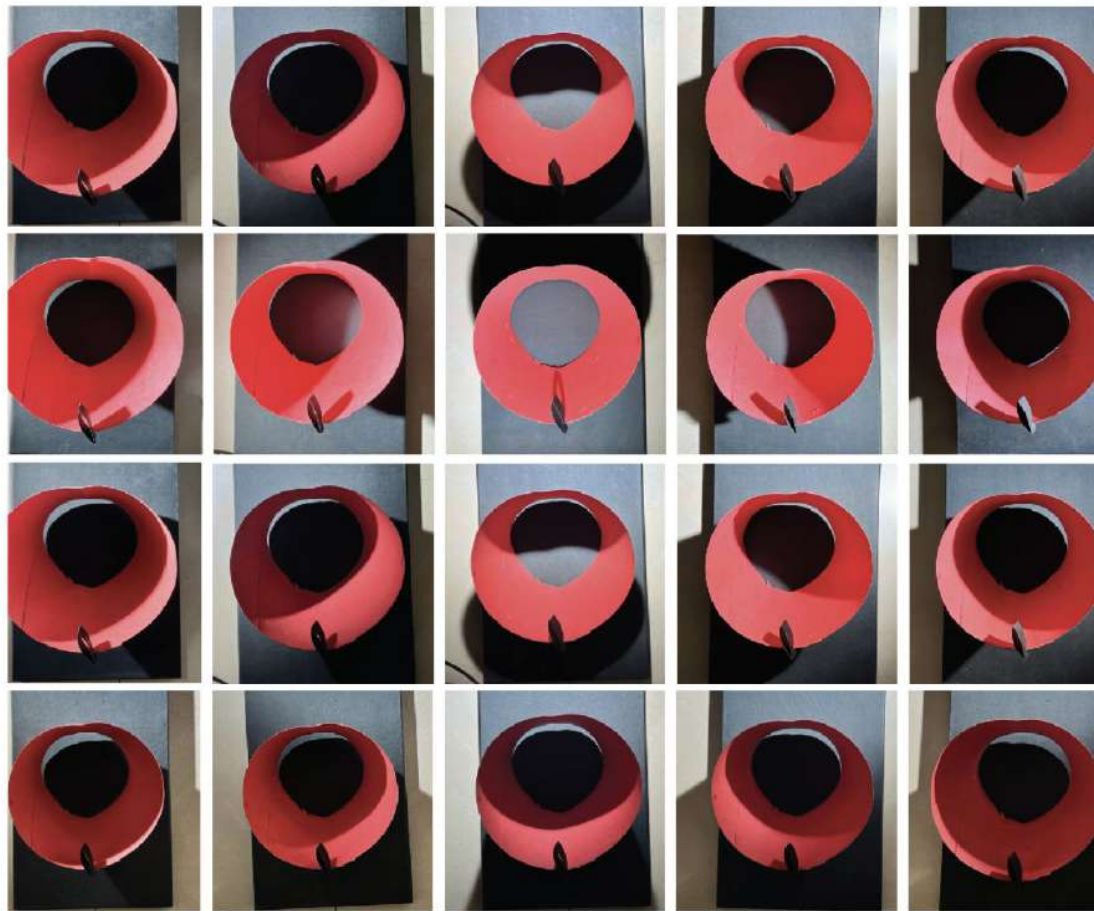




Physical model built by the team to test how the shadows will form





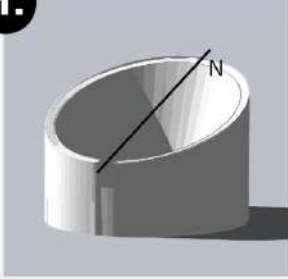


Paper model experiments mapping the shadows on the face of the building

Reasons why the sundial idea can work

Certain properties of the building make the sundial idea quite viable.

1.



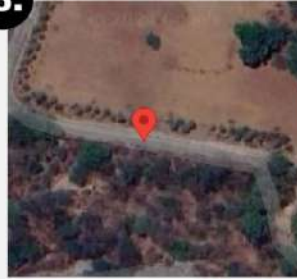
The 'truncation' of the cylinder building is aligned closely to True North. This means that the shadows formed will be symmetrical - just like a well positioned sundial.

2.



The site is about 6 kms away from the Tropic of Cancer. Which means it gets a lot of sun, but also dynamic light and shadow variations and a particularly high solar angle at noon (almost perpendicular), symbolising 1200 hours.

3.



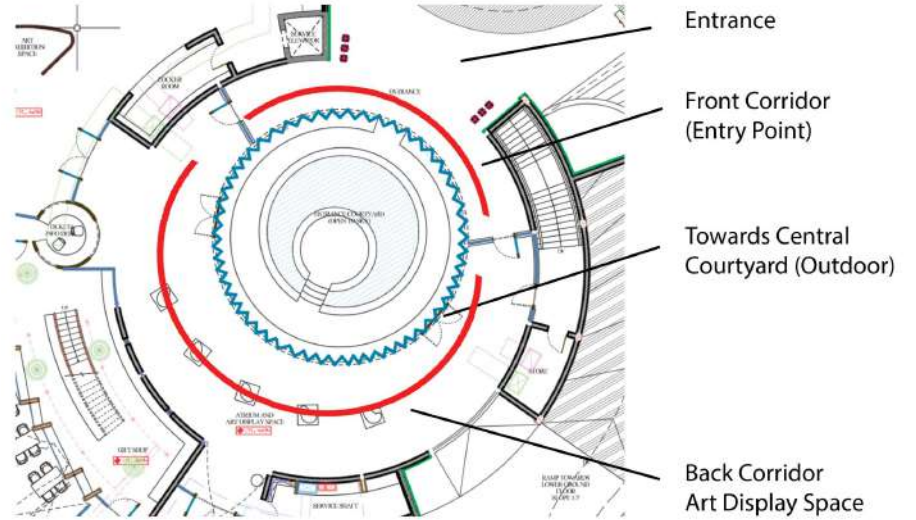
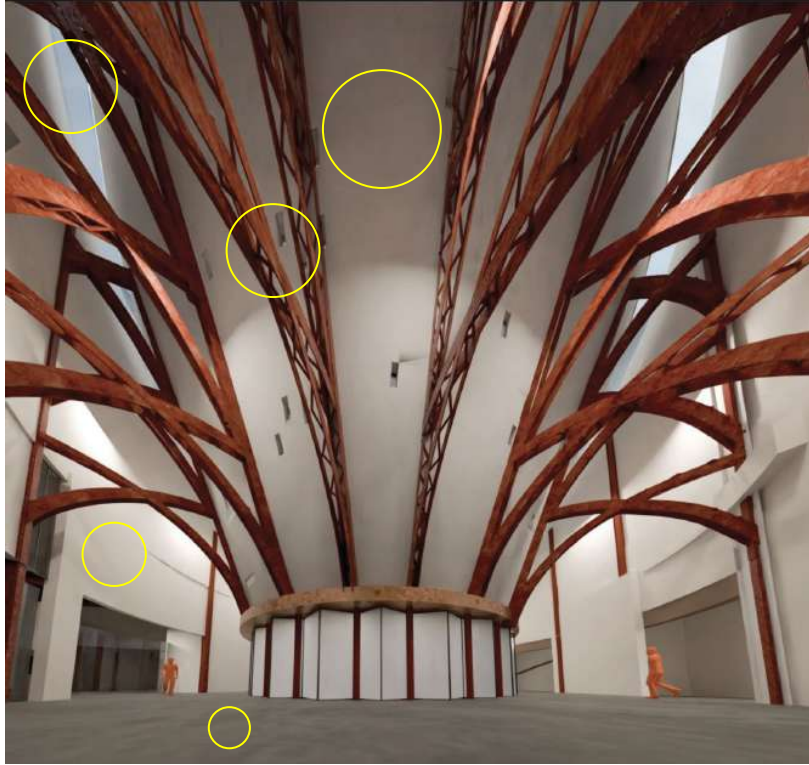
The site is located in a foresty area without any tall buildings around. This saves the sundial from any collateral shadow interferences.

4.

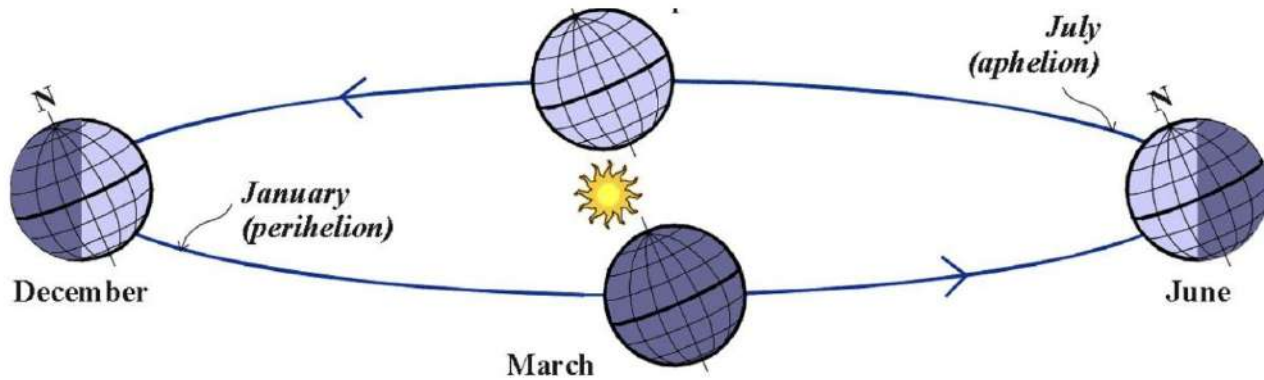


The remoteness of the site also means that it is less likely to have any light or dust pollution, which should help reduce the refraction of sunlight from the atmosphere (this prevents inaccuracies), and create crisp, sharp shadow lines.



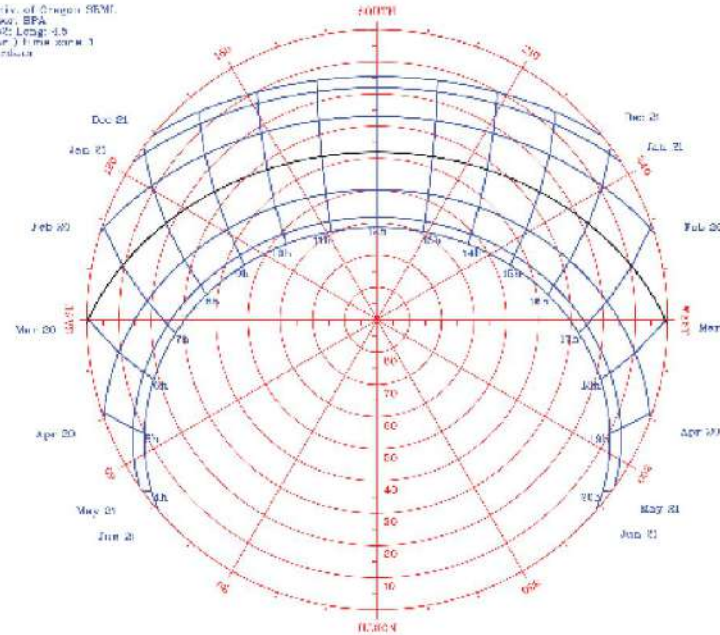


Hand-drawn architectural sketches of a building facade and interior. The sketches include a side elevation with a 22° angle, a perspective view of a room with a grid ceiling, and a plan view. Annotations include '22°', '298', '298', '12.34', 'North', 'Perspective', 'Section', and 'Plan'.

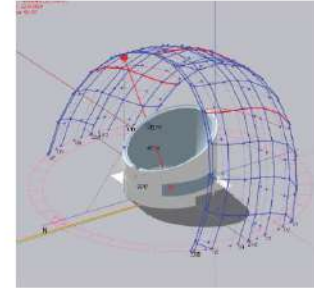
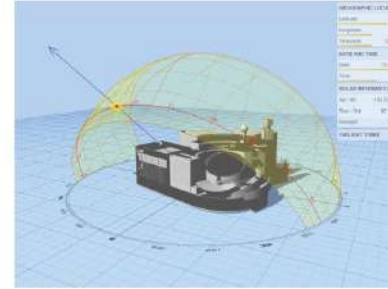


Solar Chart

Univ. of Oregon SEMI.
name: SFA
lat: 52.15deg
lon: 101.15deg
date: 1/1/2000
time: 12:00

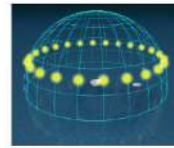
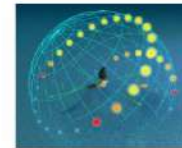
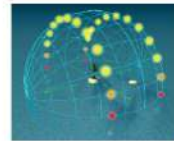


Polar sun path chart program, University of Oregon



This is a solar chart, it's a tool that helps us understand, or map the movement of the sun throughout the year by visualising it's location in the sky. It creates an arched grid by dividing the sun's movement by days (curved horizontal lines) and hours, (curved vertical lines). All in all, it is just a simple way to map out all the "possible" locations of the sun for a given place on Earth.

The chart obviously is different for different locations on Earth. For example, places near the north pole have a lower arch.



https://en.wikipedia.org/wiki/Sun_path



Shadow Mapping

	7:30 AM	9:45 AM	12:00 PM	2:45 PM	5:30 PM
20th March (Vernal Equinox)					
21st June (Summer Solstice)					
22nd September (Autumnal Equinox)					
21st December (Winter Solstice)					

7:30

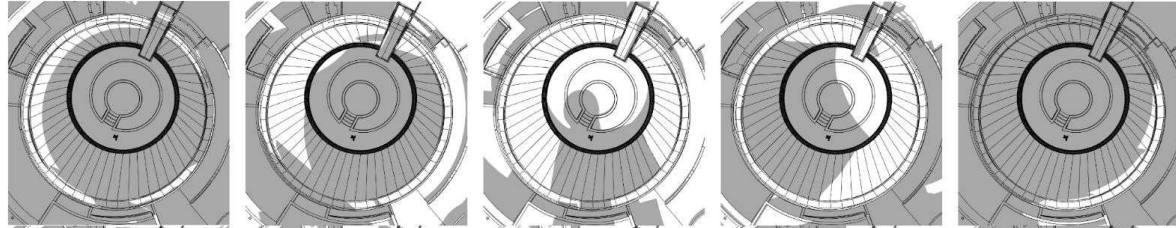
9:45

12:00

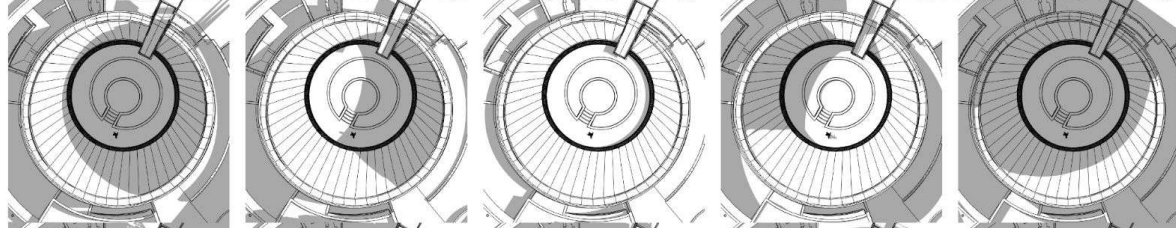
2:45

5:30

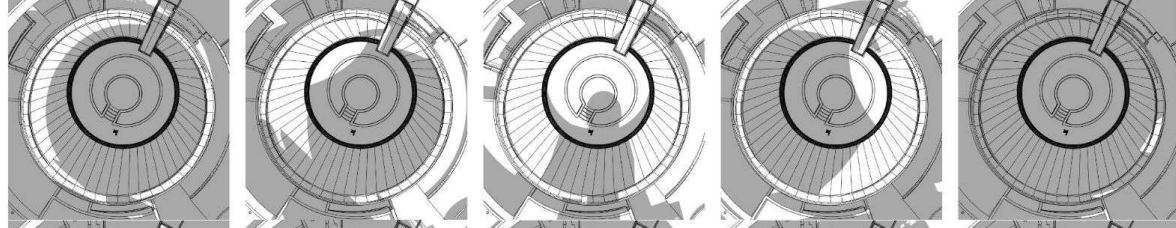
Vernal Equinox



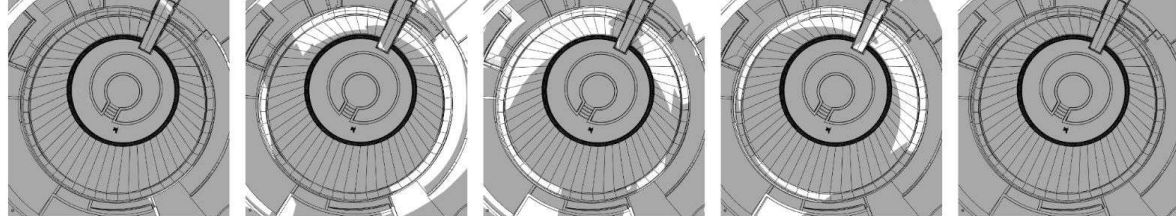
Summer Solstice



Autumnal Equinox



Winter Solstice



7:30

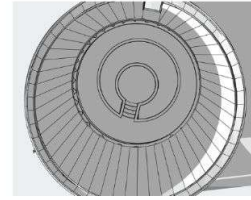
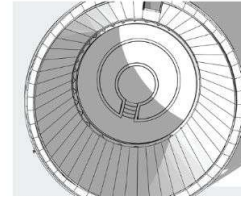
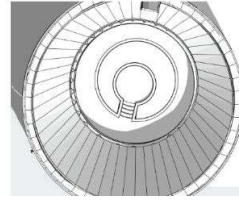
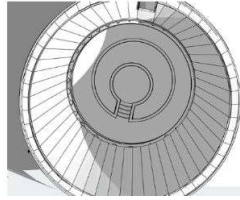
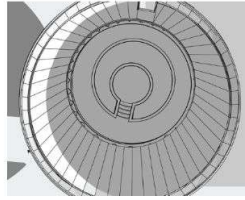
9:45

12:00

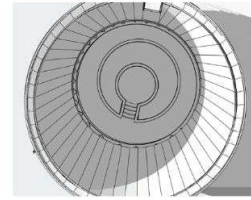
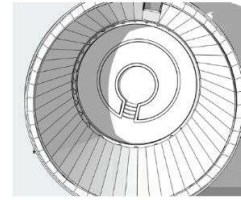
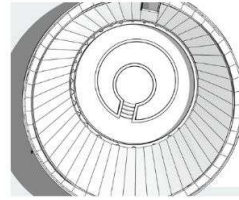
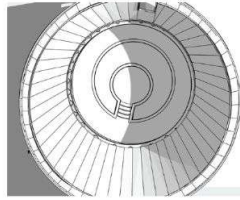
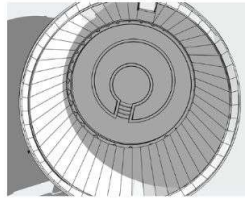
2:45

5:30

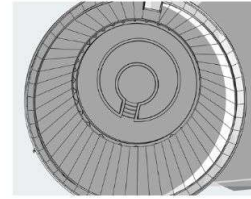
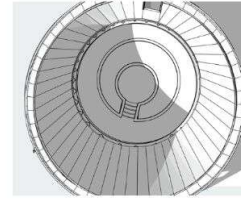
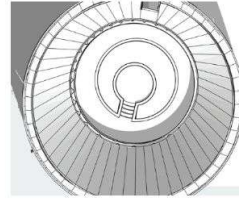
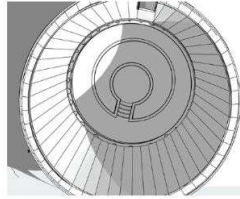
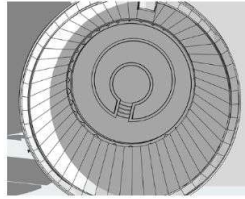
Vernal Equinox



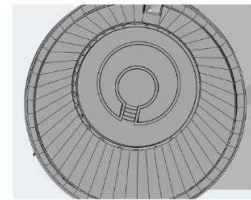
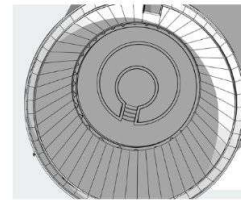
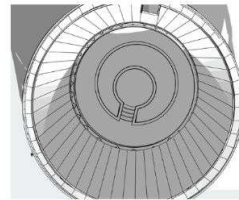
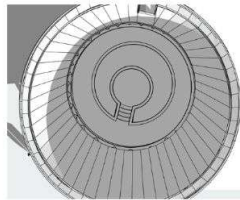
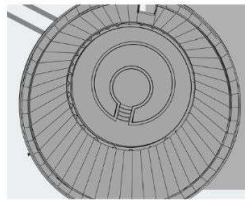
Summer Solstice



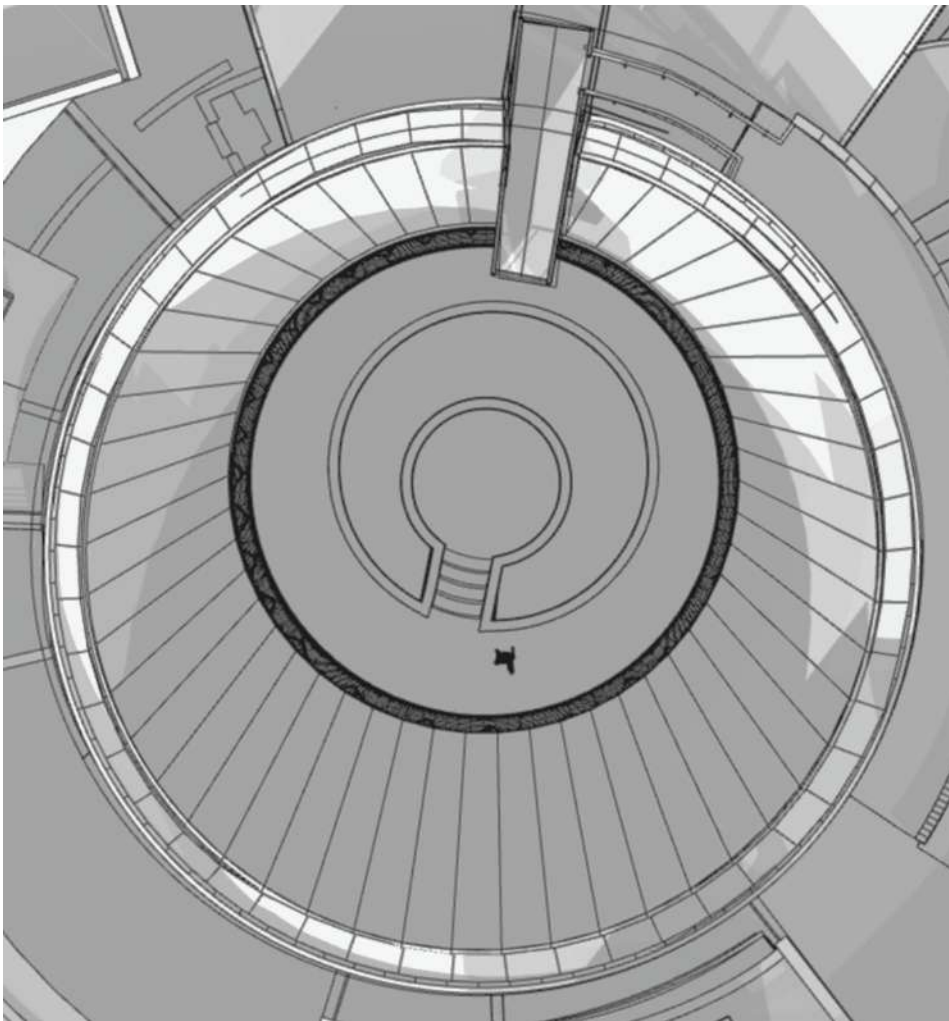
Autumnal Equinox



Winter Solstice

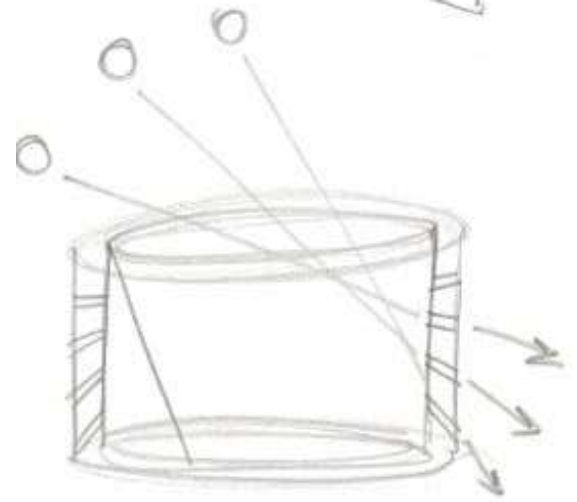
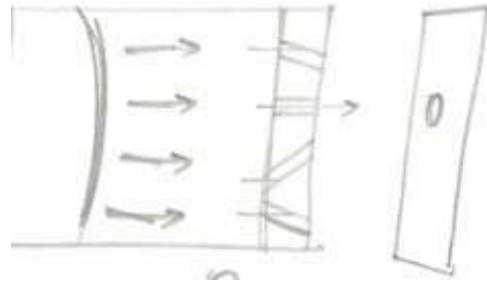
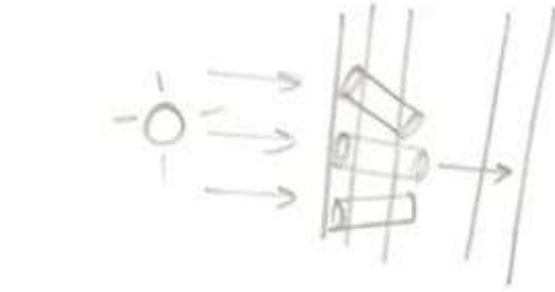
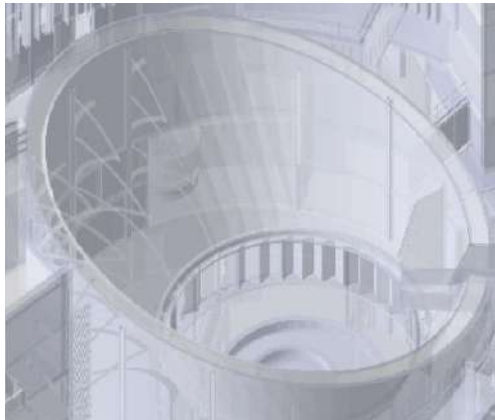
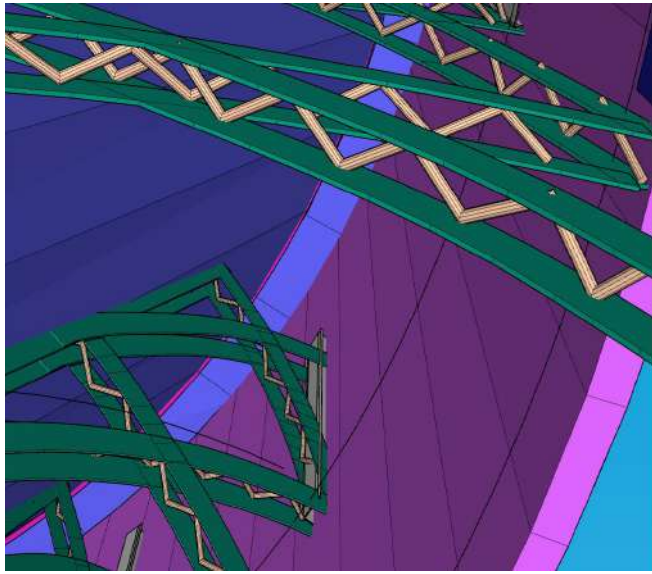


A clear symmetrical pattern comes through, highlighting the usefulness of aligning the funnel with True North. The slight asymmetry is because of the 7° offset from True North.

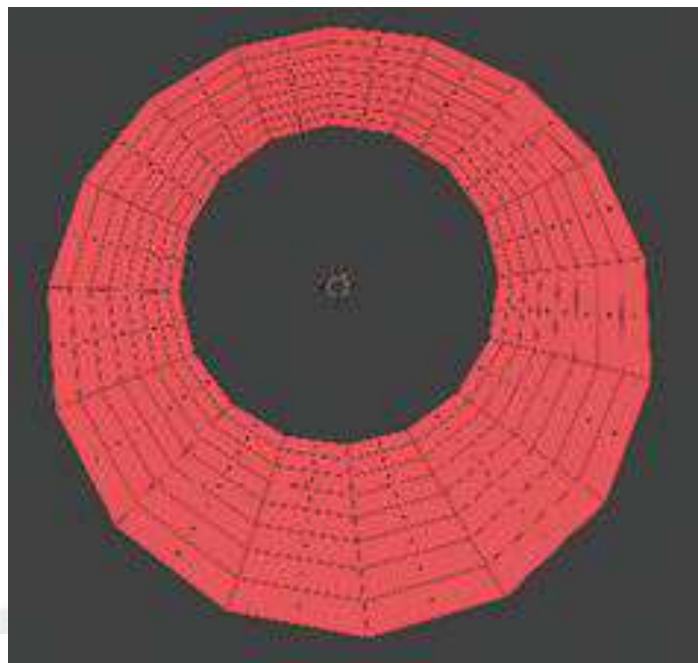
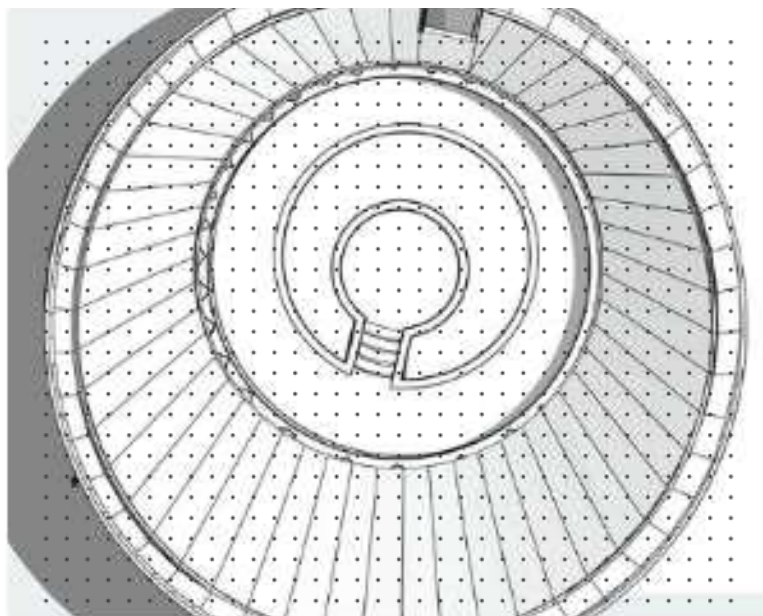


Findings

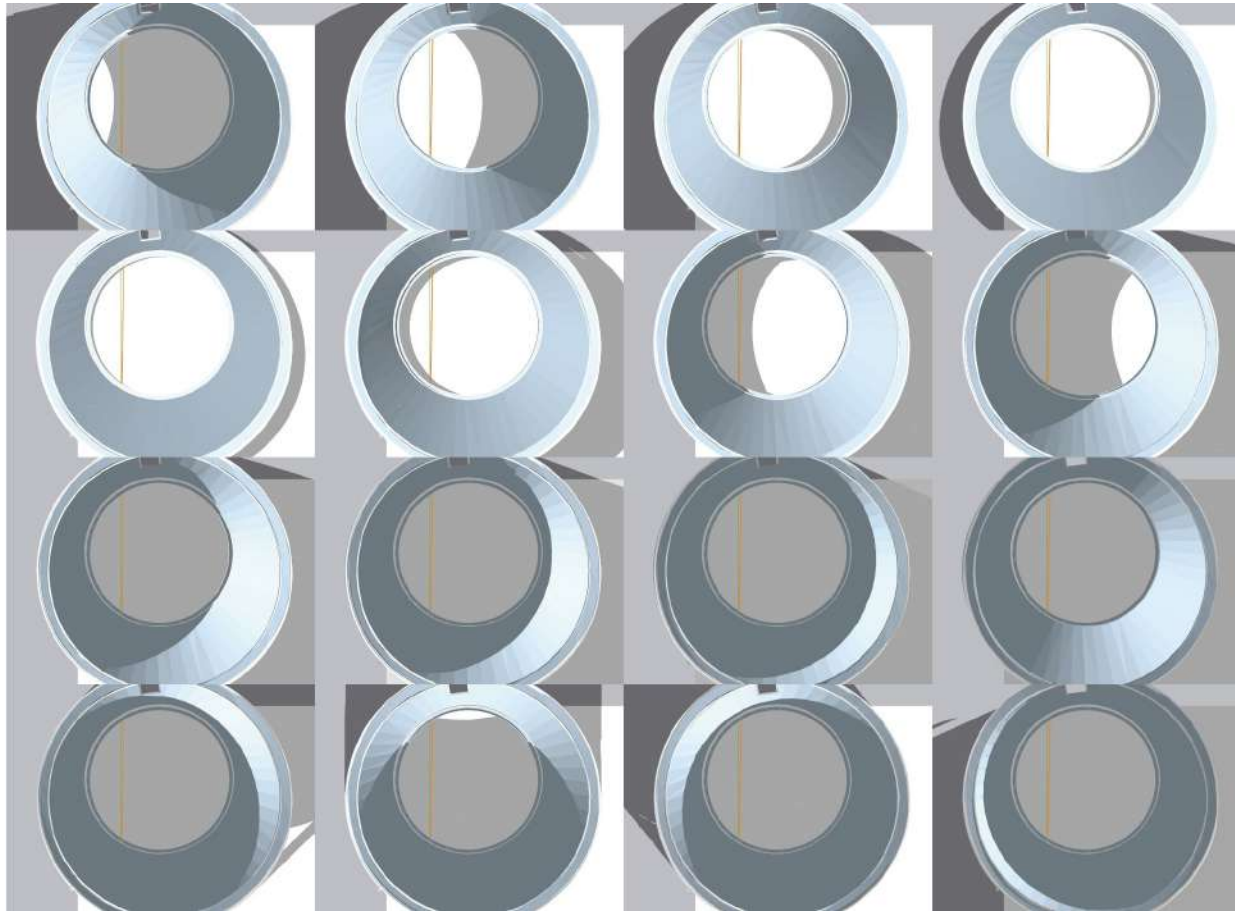
1. The biggest thing we couldn't consider was that the shadows of the buildings around the cone would disrupt the symmetry of these ordered shadow patterns. So a clear way to make marking was going to be more challenging now.
2. There is definitely lesser light on the south side of the building. It appears that it's usually dark except at noon on most days.
3. We see that the patterns of light formed on Vernal and Autumnal equinox are identical. So technically there are only 6 months that are unique, after that the sun cycle just goes in reverse.
4. There is definitely more light higher up in the building. Light only reaches on the courtyard or low walls near noon.
5. The shadows projected by the viewing deck isn't that significant, or clean. It would be unideal to utilise it as a gnomon.
6. The shadows will most likely be hard shadows. But the site is probably going to be so bright that even areas which don't see to get light will get ample ambient light. But of course, the areas with direct sunlight will be very bright.



Angled Perforations



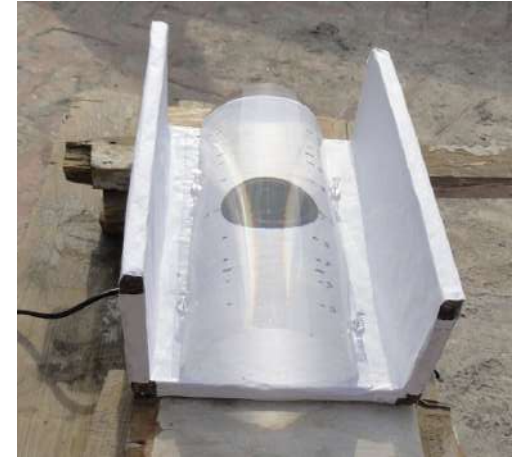
Mapping shadows on a grid



The four diagrams illustrate the rotation of a circular structure. Each diagram shows a circle with a central spiral and a radial line. The radial line is divided into segments, and the segments are numbered 1 through 12. The diagrams show the radial line rotating clockwise, with the segments moving from their initial position to a new position. The segments are labeled with numbers 1 through 12, and the radial line is labeled with 'R'.

The four diagrams show a person moving in a circular room. In the first, the person is at the top. In the second, they move clockwise. In the third, they continue clockwise. In the fourth, they are at the bottom, having completed a semi-circle. This illustrates the concept of a 'path' as a sequence of positions over time.

The four diagrams illustrate the rotation of a circular structure. Each diagram shows a circle with a central spiral and a radial line. The radial line is labeled with '1' at the top and '2' at the bottom. The spiral is labeled with '1' at the top and '2' at the bottom. The diagrams show the radial line and spiral rotating together, with the radial line always pointing towards the top of the circle.



SOLAR TUBE – INNOVATIVE DAYLIGHTING

CHALLENGE OF BEING UNIQUE, ENERGY CONSCIENCE, VALUE ENGINEERING, FLEXIBILITY – REFERENCE
IMAGE

UBE

1. LIGHT COLLECTING:

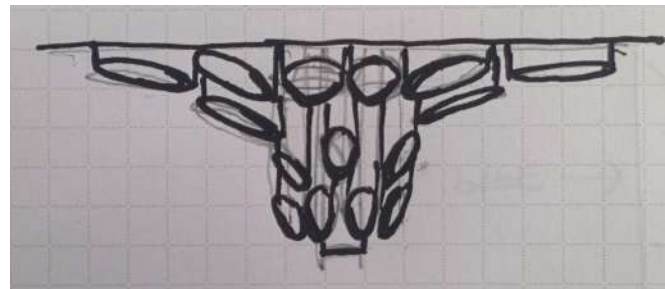
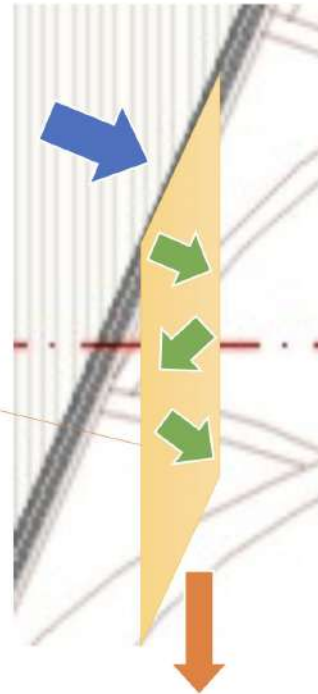
A special lens efficiently captures and redirects sunlight

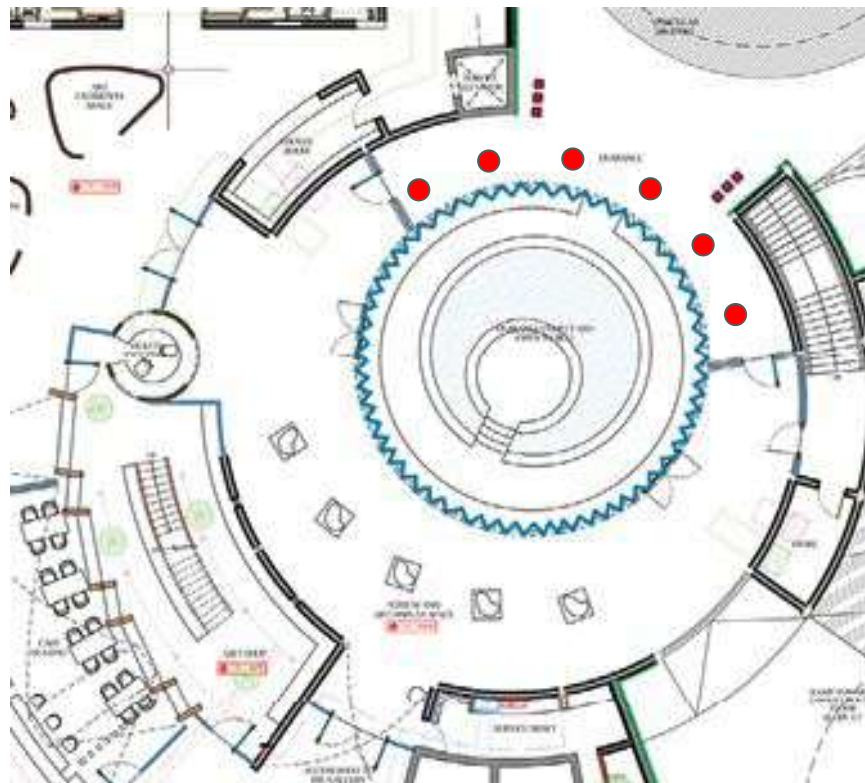
1. LIGHT TRANSITING:

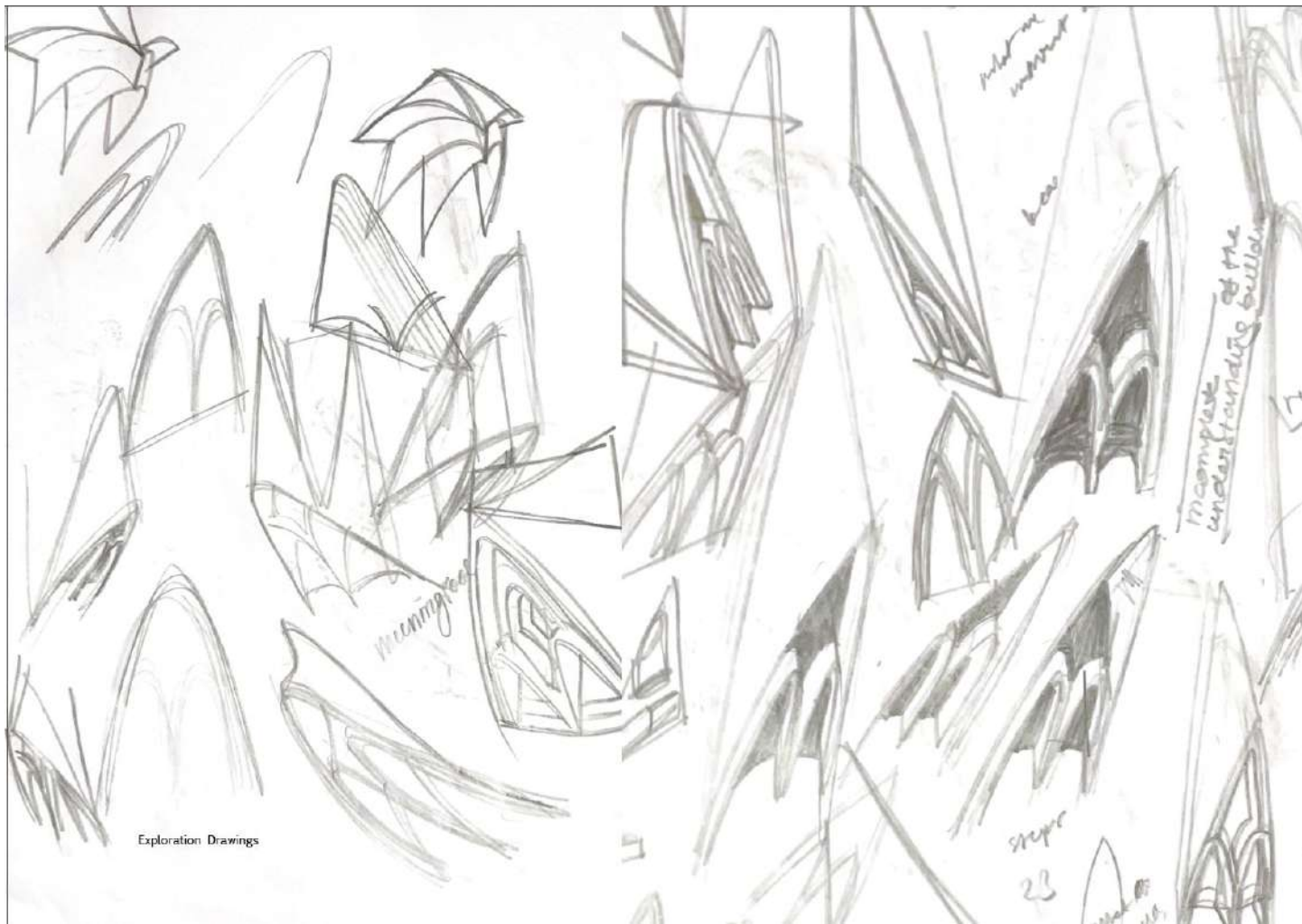
The light travels through a highly reflective *aluminium plated tube*

1. LIGHT DIFFUSING:

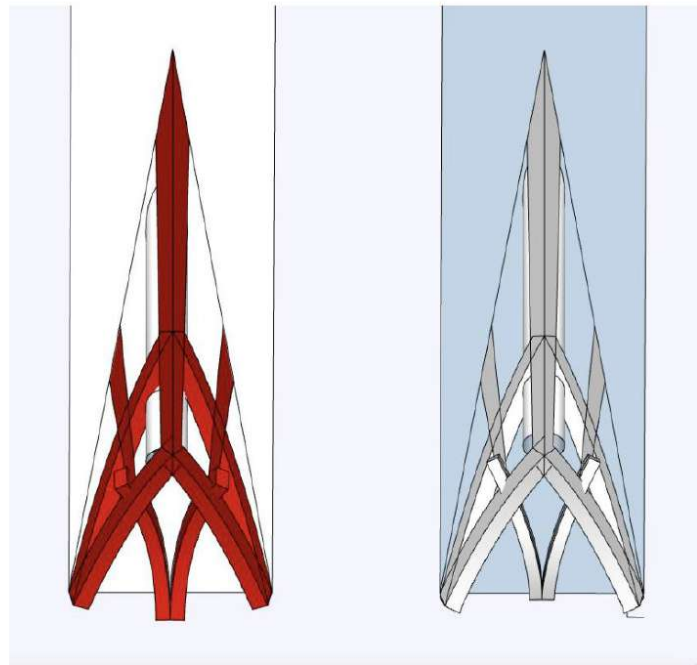
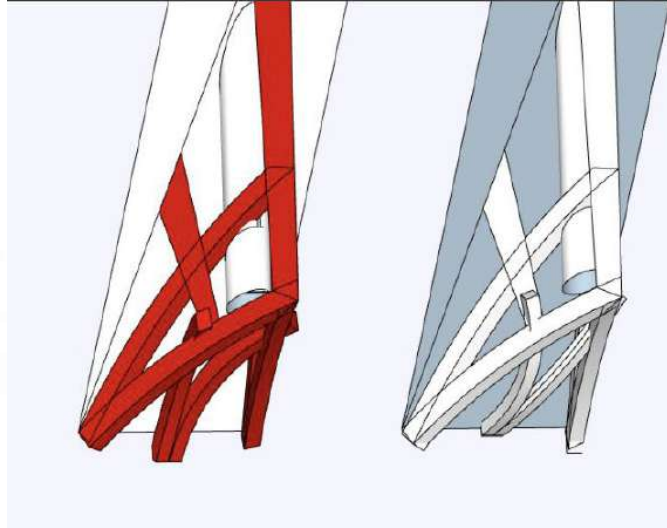
The Light hits a diffuser and creates illumination

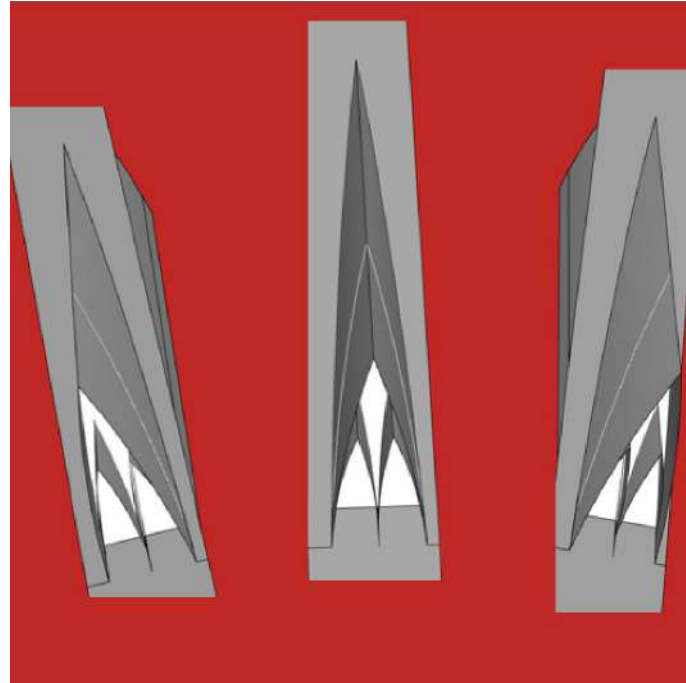
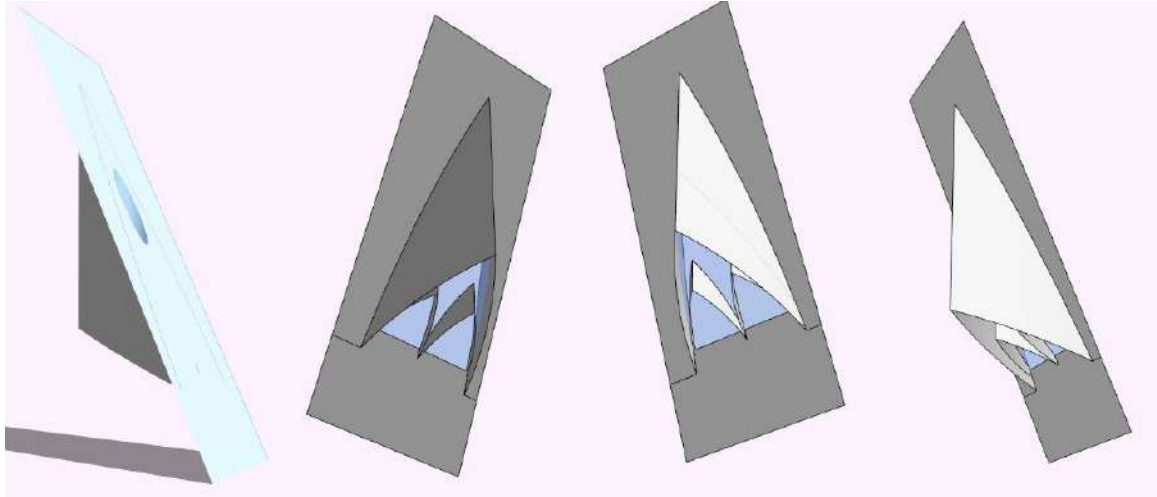
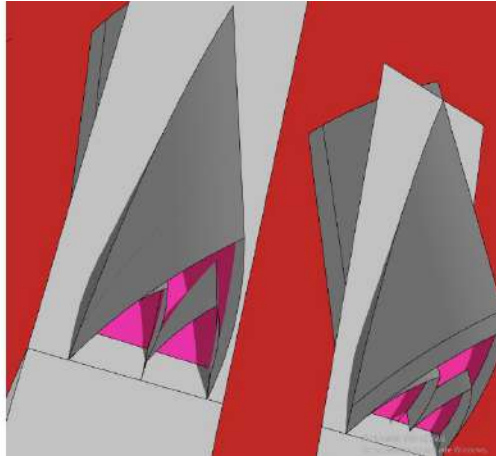
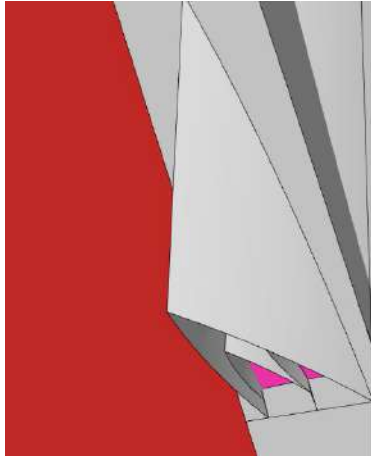


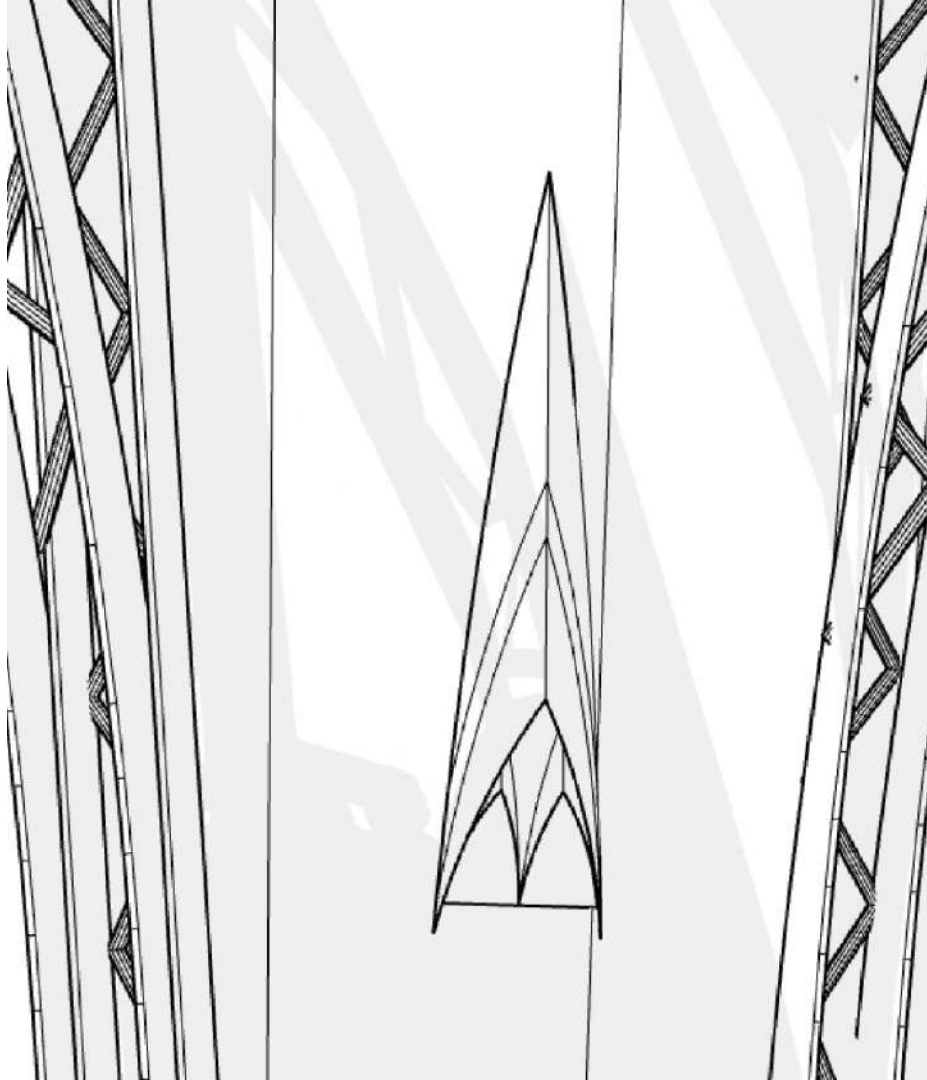


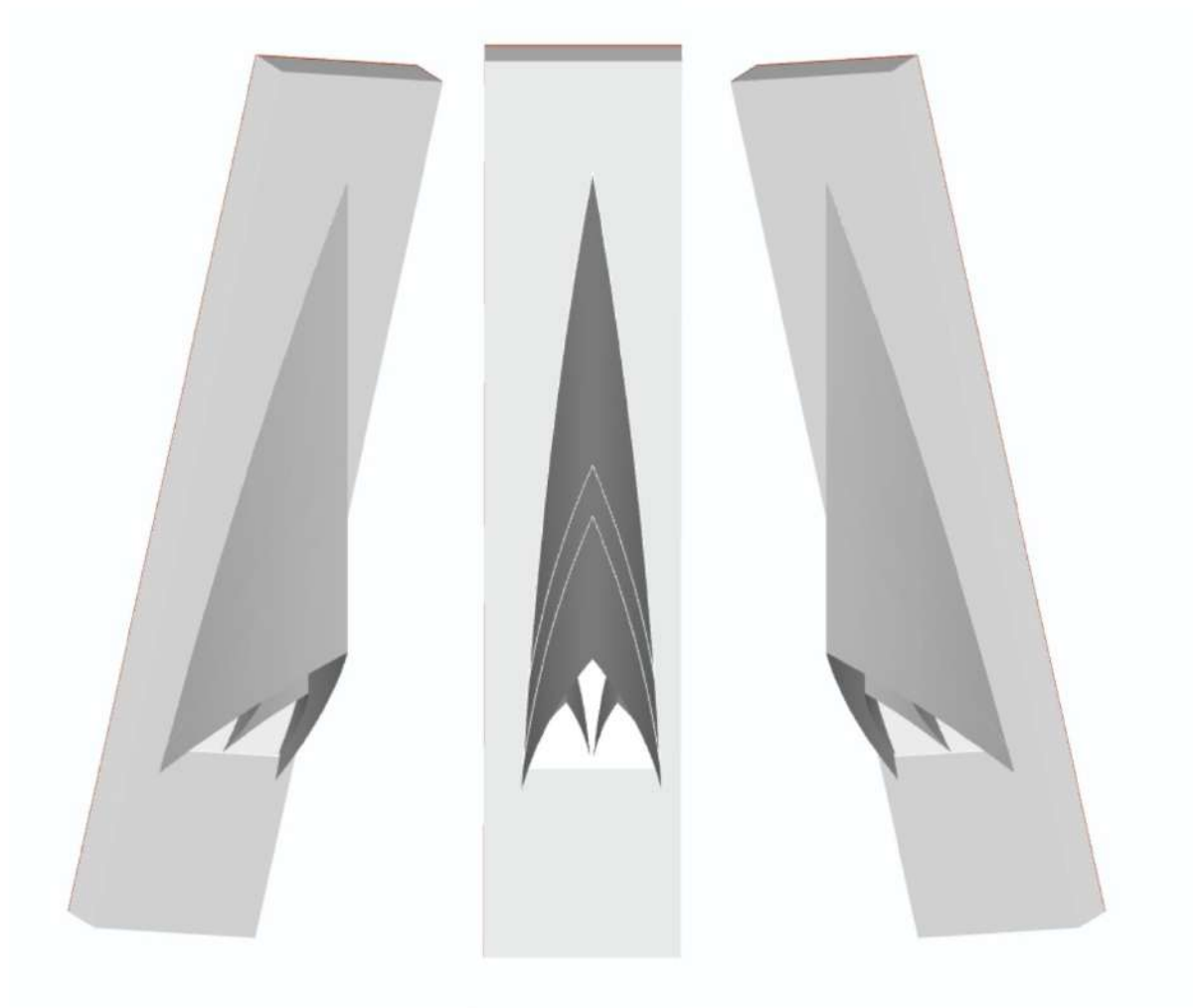


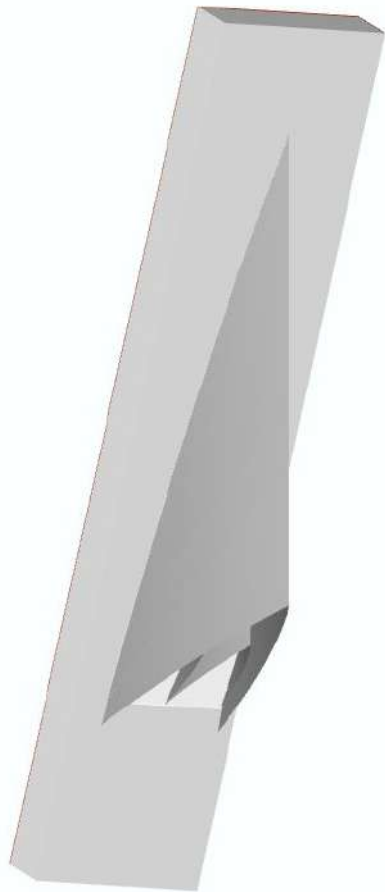
Exploration Drawings

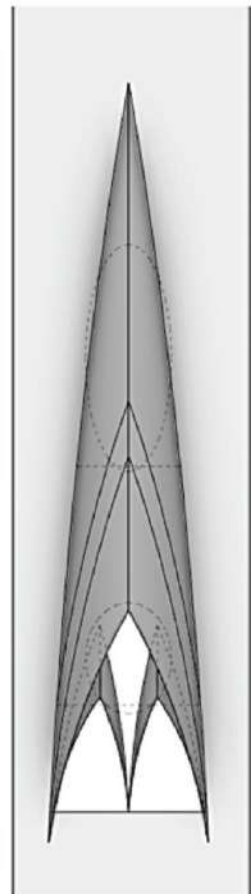
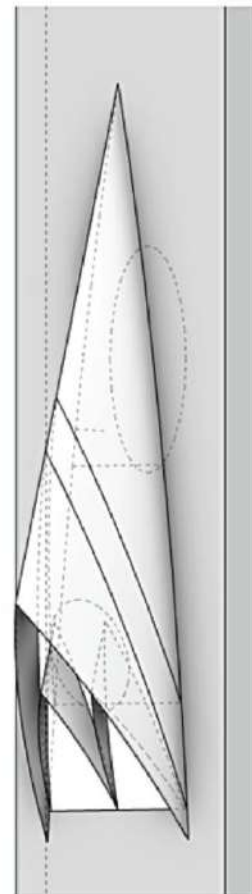
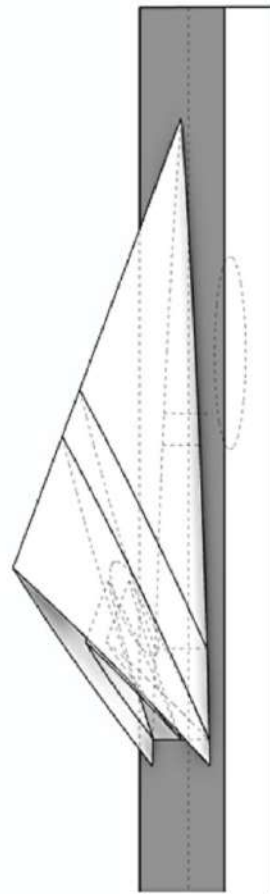
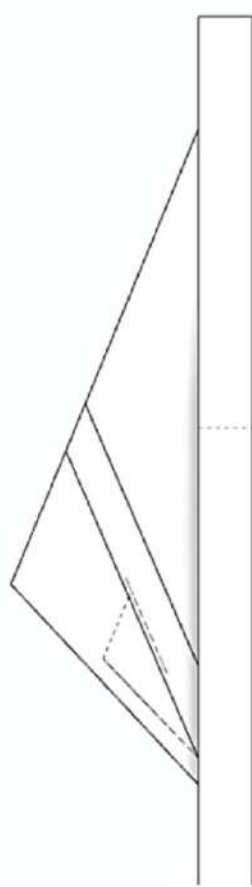
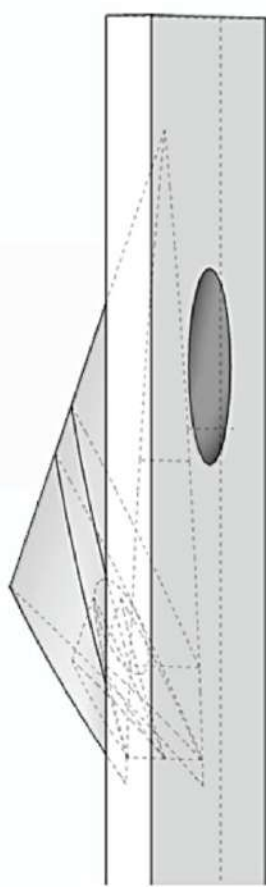
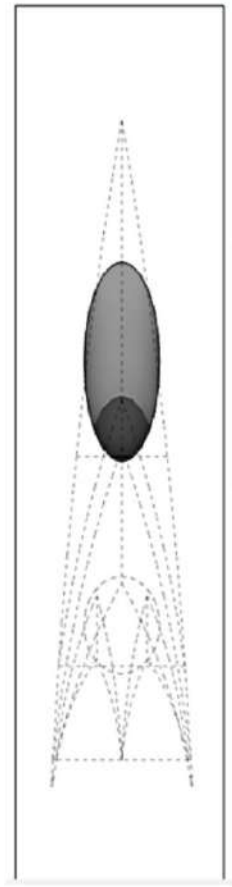












Facade

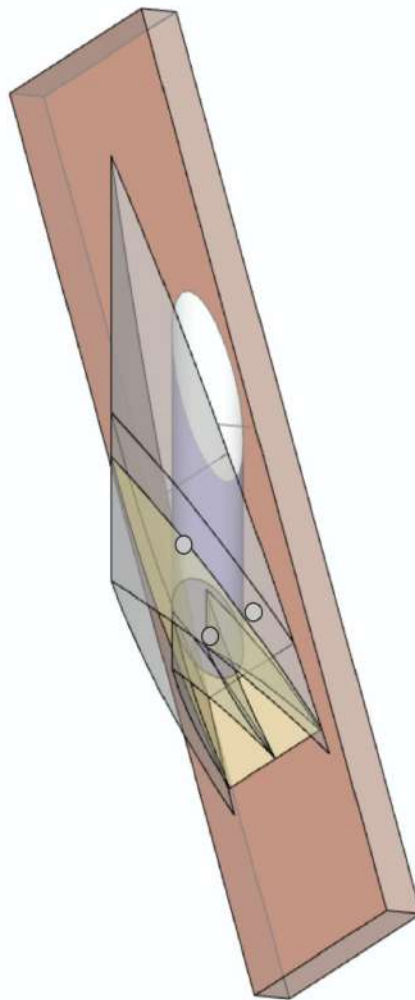
Lens

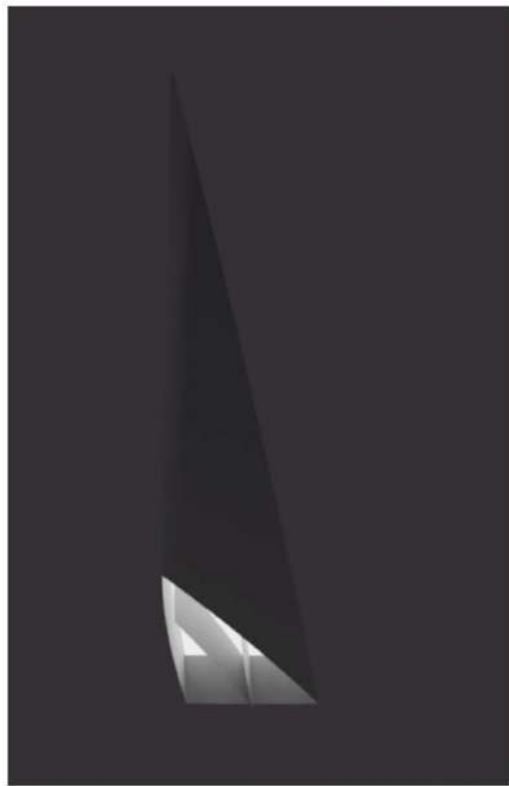
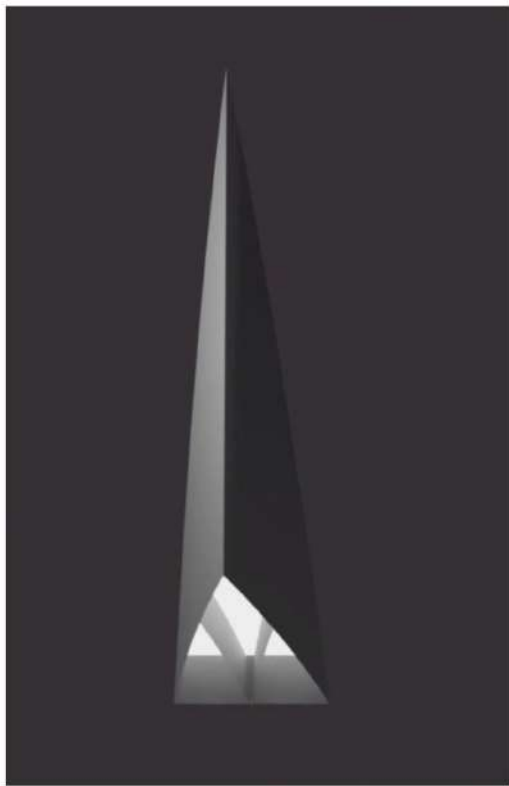
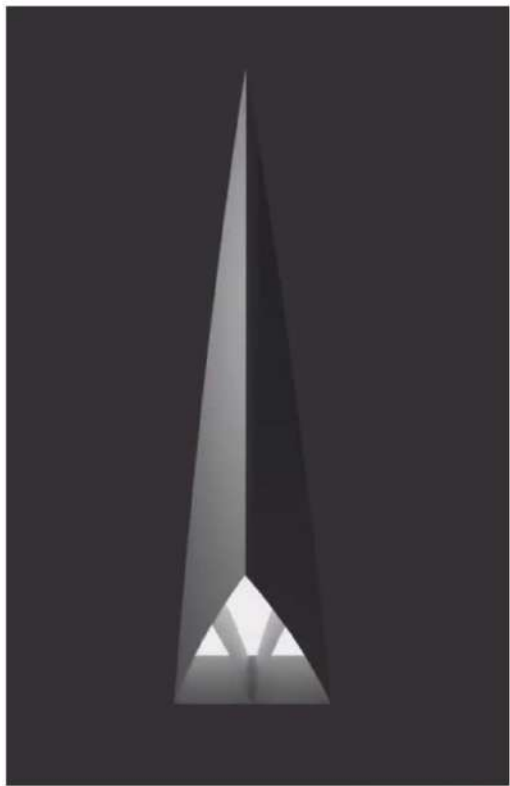
Reflective Tube

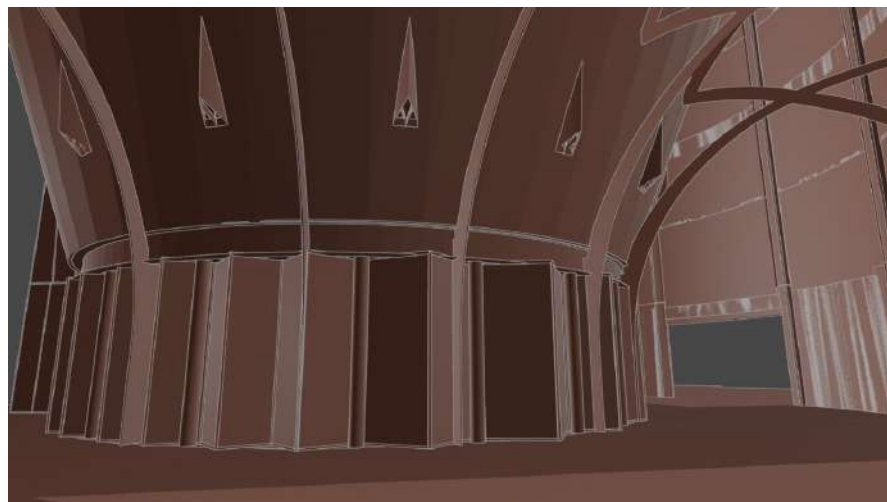
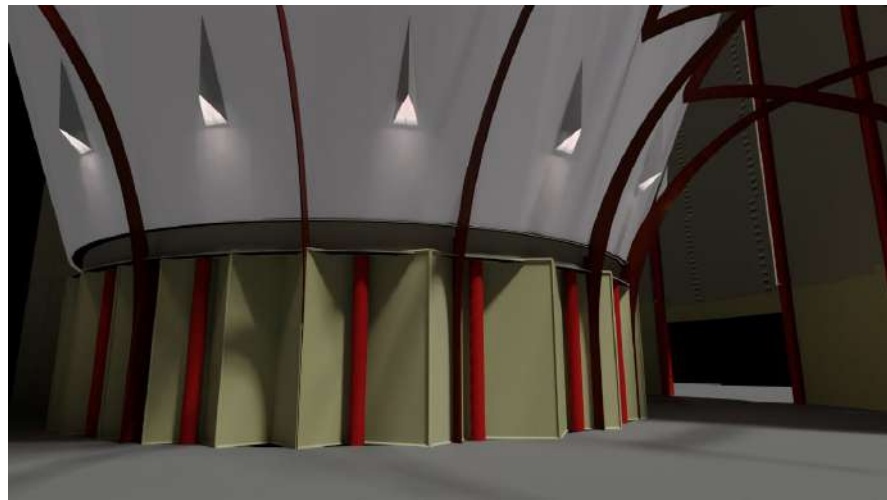
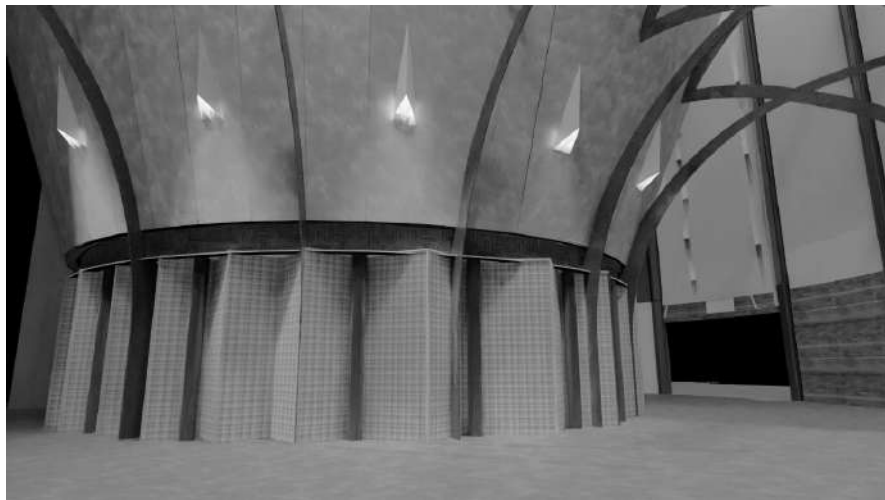
Panel 1 (PCB)

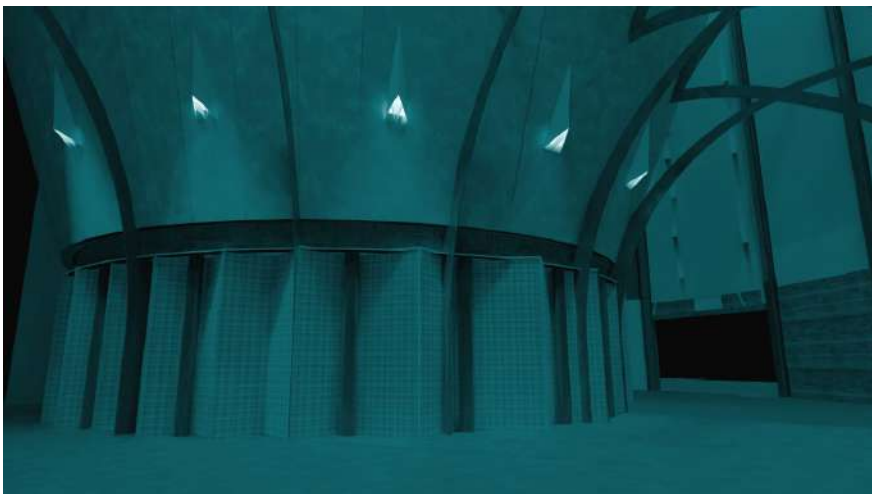
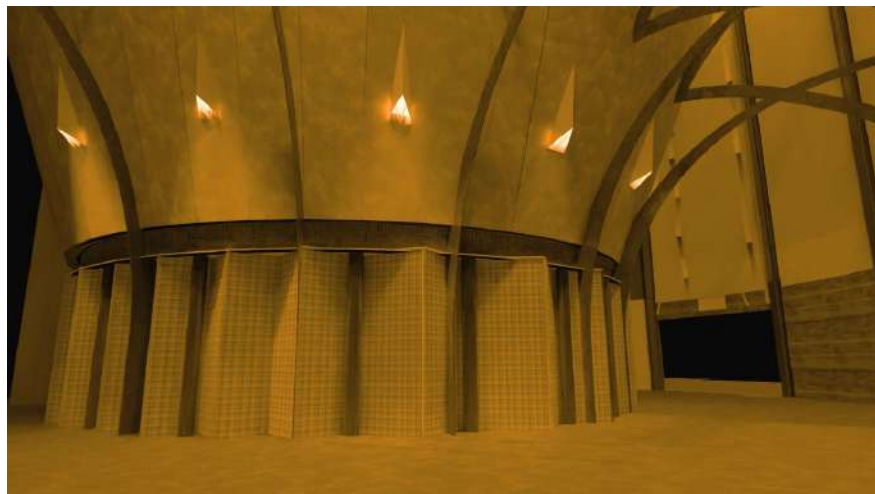
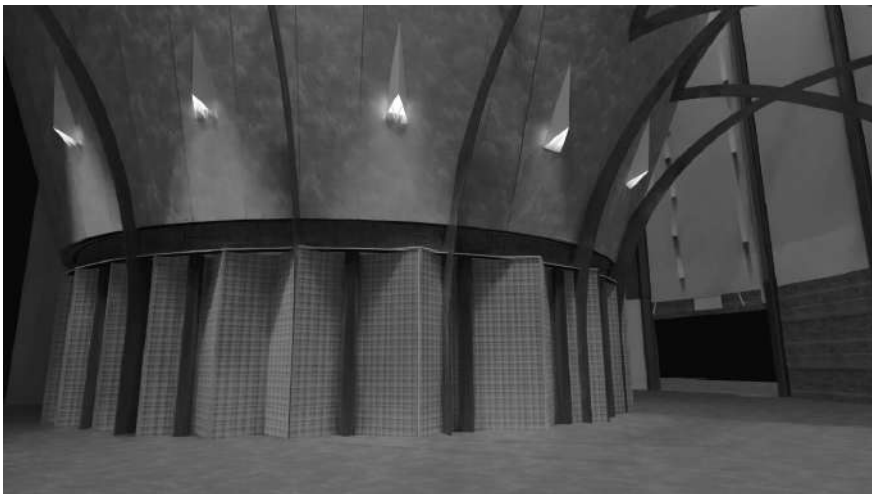
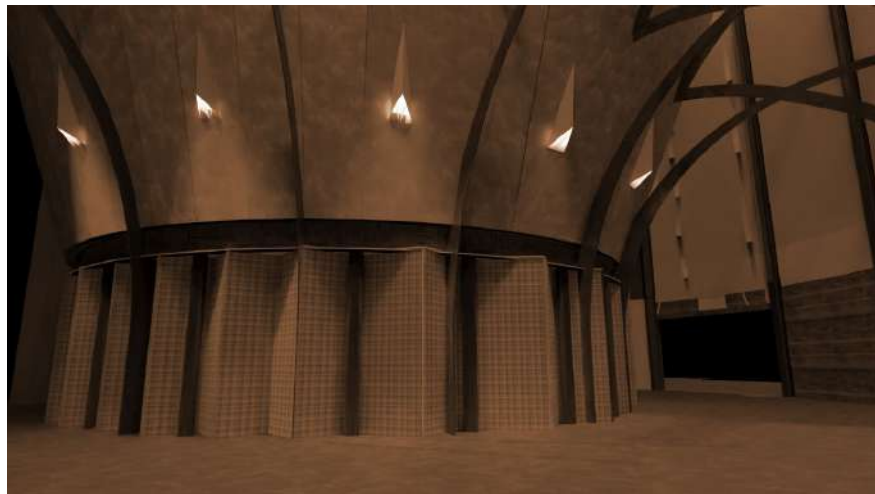
Frosted PMMA

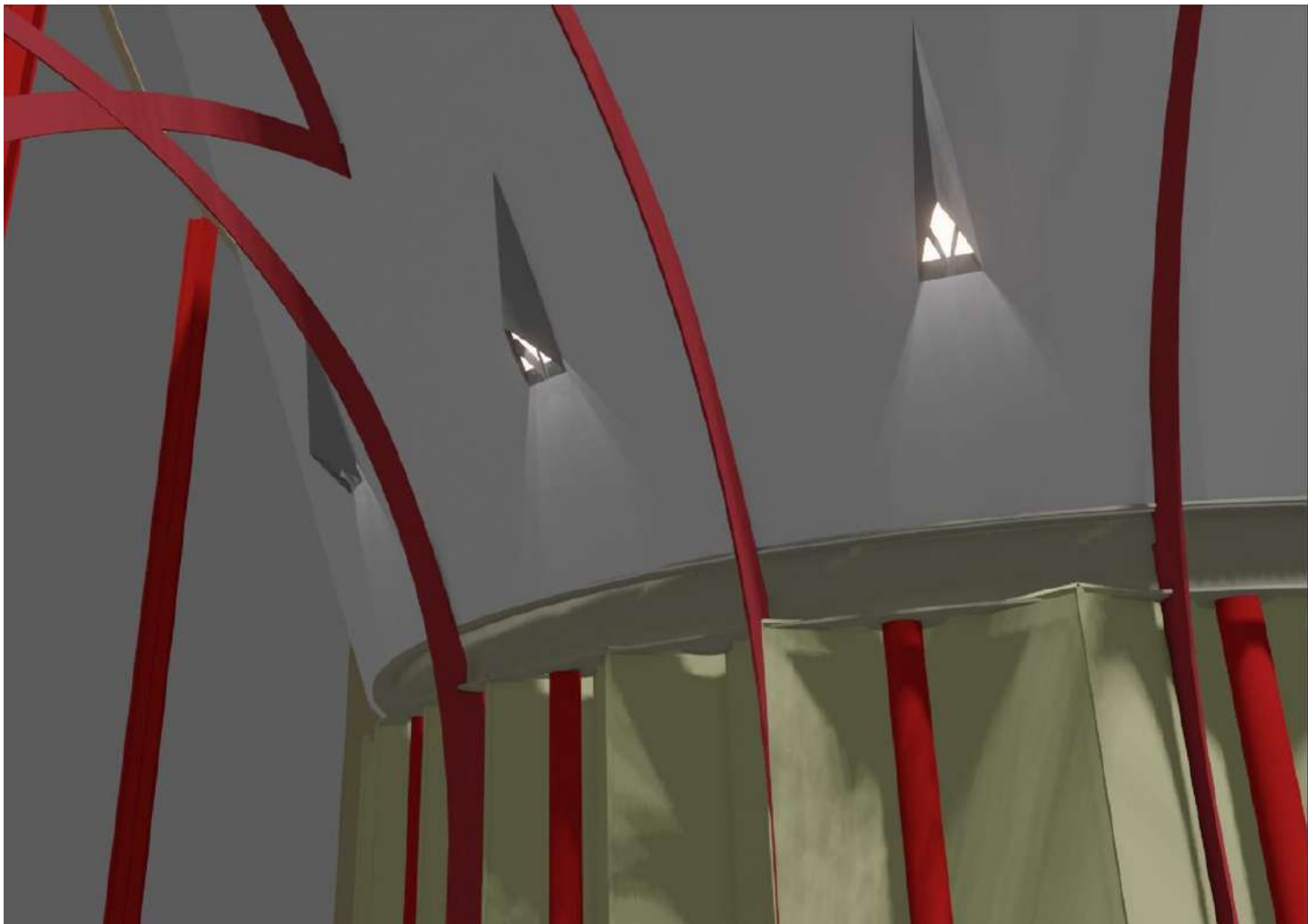
Aluminium Fitting











Solar Pipe Lighting Fixture, Final
Proposed Design (Approved)